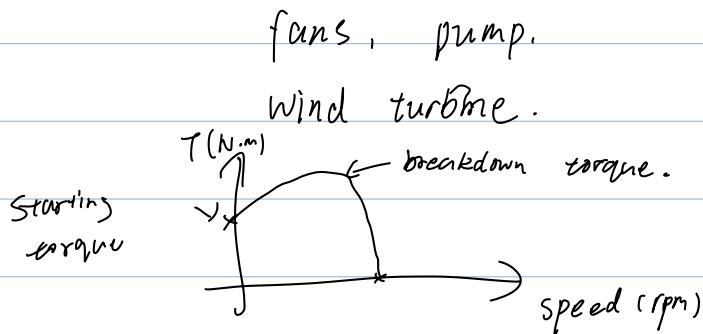
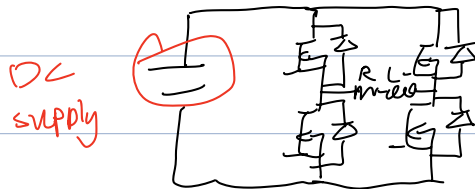


2023.

Q1. (a).



(b).



switch diagonal pairs to apply $+V_{dc}$ and $-V_{dc}$
to the load RL .



Difference between fixed duty PWM and sin PWM
control :

fixed duty: duty ratio is constant
→ more harmonics
→ poor waveform quality

sinusoidal: duty follows a sinusoidal reference
→ lower harmonics.

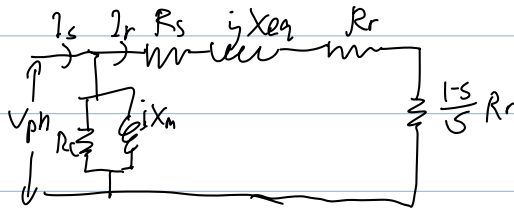
→ fundamental voltage magnitude controlled by M_a .

Non sinusoidal load current leads to increased losses in motor.

$$(c), \text{ from } N_s = \frac{120f}{p} \Rightarrow f_s = \frac{p}{120} = \frac{4 \times 500}{120} = 50 \text{ Hz}$$

$$\text{From } V_{L_{rms}} = 0.612 V_{dc} \cdot M_a \Rightarrow M_a = \frac{V_{L_{rms}}}{0.612 \times V_{dc}} = \frac{200}{0.612 \times 400} = 0.817$$

$$(d) \quad s = \frac{N_b - N_r}{N_s} = \frac{1500 - 1350}{1500} = 0.1$$



$$\frac{1-s}{s} R_r = \frac{1-0.1}{0.1} \times 0.2 = 1.8 \Omega$$

$$Z_T = R_s + R_r + \frac{1-s}{s} R_r + jX_{eq} = 0.2 + 0.2 + 1.8 + j \cdot 0.6 = 2.280 \angle 15.25^\circ \Omega$$

$$|I_r| = \frac{V_{ph}}{|Z_T|} = \frac{200/\sqrt{3}}{2.280} = 50.645 \text{ A}$$

$$P_{mech} = 3 |I_r|^2 \cdot \frac{1-s}{s} R_r = \boxed{13850.5 \text{ W}}$$

$$T = \frac{P_{mech}}{\omega} = \frac{13850.5}{1350 \times \frac{2\pi}{60}} = \boxed{97.972 \text{ N}\cdot\text{m}}$$

$$P_{loss} = 3 |I_r|^2 \cdot (R_r + R_s) + 3 \cdot \frac{V_{ph}^2}{R_c} = \boxed{3744.6 \text{ W}}$$

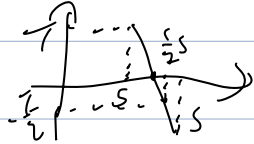
$$\text{Efficiency} = \frac{P_{mech}}{P_{mech} + P_{loss}} \times 100\% = \boxed{78.72\%}$$

$$(e) \quad s' = 0.05$$

$$(i). \Delta E = \frac{1}{2} C (V_{440}^2 - V_{400}^2) = 336 \text{ J}$$

$$|T'| = \frac{T}{2} = 48.986 \text{ N.m}$$

(Assume the T_{ω} speed curve is symmetrical).



$$\text{From } s = \frac{N_s - N_r}{N_s} \Rightarrow N_r = 1425 \text{ rpm} \Rightarrow N_r = 1575 \text{ rpm}$$

$$P_{\text{mech}} = |T'| \cdot \omega' = 7310.0 \text{ W} \rightarrow 8079.4 \text{ W}$$

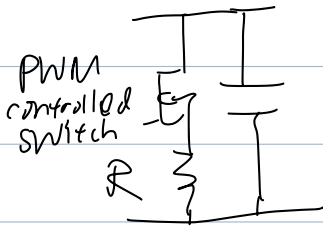
Assume the efficiency is still 78.72% :

$$P_{\text{dc}} = P_{\text{mech}} \cdot \text{efficiency} = 5717.9 \text{ W}$$

$$t = \frac{\Delta E}{P_{\text{dc}}} = 58.763 \text{ ms}$$

Assume inverter efficiency is 98% $\rightarrow P_{\text{dc}} = 6232.9 \text{ W}$

(ii)



$$(iii). t = \frac{1}{f_{\text{switch}}} = 0.1 \text{ s} = 100 \text{ ms}$$

$$\Delta E = \frac{1}{2} C (V_{450}^2 - V_{470}^2) = 176 \text{ J}$$

$$t_{\text{rise}} = \frac{\Delta E}{P_{\text{dc}}} = 27.67 \text{ ms}$$

$$\Rightarrow t_{\text{fall}}: t_{\text{fall}} + t_{\text{rise}} = 72.33 \text{ ms}$$

$$P_{dc} = \frac{\Delta E}{t_{\text{fall}}} = 2433.29 \text{ W}$$

$$P_{\text{from}}: P_{dc} = P_{\text{brake}} - P_{dc}$$

$$\begin{aligned} \Rightarrow P_{\text{brake}} &= P_{dc} + P_{dc} \\ &= 2433.29 + 6360.1 \\ &= 8793.4 \text{ W} \end{aligned}$$

$$P_{\text{from}} P_{\text{brake}} = \frac{V_{\text{AUG}}^2}{R_{\text{brake}}}$$

$$\Rightarrow R_{\text{brake}} = \frac{V_{\text{AUG}}^2}{P_{\text{brake}}} = \frac{440^2}{8793.4} = 22.017 \Omega$$

$$[f]_1: P_{\text{mech}} = \frac{1}{2} \rho \pi R^2 C_p v^3$$

$$30 \text{ kW} = \frac{1}{2} \cdot 1.225 \cdot \pi \cdot 8^2 \cdot 10^3 C_p$$

$$\Rightarrow C_p = 0.2436$$

$$TSR = 5 = \frac{WR}{V} \Rightarrow \omega = 6.25 \text{ rad/s}$$

$$N_r = N \omega \cdot \frac{60}{2\pi} = 1790.49 \text{ rpm}$$

$$N_s \approx N_r = \frac{120 f_s}{p} \Rightarrow f_s = \frac{p N_r}{120} = 59.68 \text{ Hz} \approx 60 \text{ Hz}$$

(e). offshore: PM synchronous generator

no gearbox, easy to maintain

and highly reliable.

onshore: Doubly-fed induction generator.

Lower capital costs and

only need converter rated at 20%

generator power.